William Chiu

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Education

University of Waterloo **2024 - Present**
Bachelor of Software Engineering, Honours Co-Op *Waterloo, ON | CGPA: 3.93/4.00*
Colonel Hugh Heasley Scholarship Recipient

Work Experience

Waterloo Vision & Image Processing Lab **May 2026 – Present**
Undergraduate Research Assistant *Waterloo, ON*

- Improved a hockey action-recognition model's **mean per-class accuracy from 44.7% → 51.8%** and **overall accuracy from 79% → 81.5%** by fusing player-motion signals (speed, direction, orientation) into a pose-only classifier; ablated **FiLM vs. linear fusion heads** across varying input volumes, finding a **hard-gated linear head** best, with gains concentrated on rare actions like backward skating.
- Validated a **skeleton-based action recognizer's** robustness to pose noise, finding $|r| < 0.04$ between keypoint confidence and action recognition accuracy across 5,383 clips from a **27K-clip NHL dataset**.

Blair AI **January 2026 – April 2026**
Software Engineer *Toronto, ON*

- Saved **7+ staff hours per week** of manual fax-filing by building a multimodal AI fax classification platform processing **60,000+ documents annually**, integrated directly into the **Avaros EMR** and deployed on **AWS** via Terraform.
- Improved **OCR prompt performance** by **35%** across **1,000+ labeled faxes** by building an **LLM-driven prompt refinement and benchmarking platform** with automated data ingestion, stratified sampling, and backtesting.
- Won **6-figure annual contracts** from major hospitals by extending the fax-processing platform into an end-to-end **MRI referral system** with a scalable multi-tenant **Postgres** model, AWS communication pipelines, and **voice AI** integration for autonomous follow-up calls.
- Built a human-in-the-loop escalation system enabling **SMS** and **voice AI** agents to route liability-sensitive cases to hospital staff and bridge **Slack threads** with real-time, two-way patient SMS conversations.

Ontario Government (MPBSDP - GSIC) **May 2025 – August 2025**
Full Stack Developer *Toronto, ON*

- Built reusable **Azure DevOps YAML pipelines** and deployed the Organ Donor Registration Platform to **Azure Kubernetes Service**, enabling automated, **zero-downtime releases** across six environments from **DEV to PROD**.
- Led development of the Organ Donor Registration backend using **Java Spring Boot**, integrating external services through **Azure API Management** and adding resilient **WebClient** retry and failure handling to improve reliability.
- Reduced email template retrieval latency by **80%** via **Caffeine caching**, and protected sensitive data for **16M+ Ontarians** through an **AOP logging** framework with a JSON data masker.
- Built 5 of 8 pages for a bilingual **Next.js/TypeScript** address-change form, implementing regex-based input validation and custom multi-step navigation that reduced form-entry errors and improved WCAG accessibility compliance.

Projects

Polyhedral Order Book | *C++20, CMake, Catch2, HiGHS*

- Implemented a **polyhedral order book** in C++20, generalizing order crossing to N assets via **primal/dual LPs** solved with HiGHS, enabling **multi-leg trades** and **correlated-basket market making**.
- Validated correctness across **crossed/uncrossed books**, **partial fills**, and **multi-level trades** against a baseline **2-asset CLOB** with a **Catch2** test suite under **CMake/Ninja**.

FitWithDyce.com | *TypeScript, NextJS, Vercel, Twilio, Resend, Supabase*

- Shipped a **full-stack fitness platform from 0 to 1** that automates client onboarding and communication via **Twilio** SMS and **Resend** email, cutting manual coordination for the coach.
- Engineered a **multi-tenant Supabase backend** with **row-level security** so each client sees only their own data, powering workout logging, coach chat, meeting scheduling, nutrition tracking, and custom messaging automation.

Quantum Computing Algorithm to Minimize Power Loss | *Python, D-Wave*

- **Minimized power loss in energy systems** through a designing a cloud-based quantum annealing **DFS**-based search strategy and earned a silver medal at the Canada Wide Science Fair.

Technical Skills

Languages: C/C++, Java, Python, JavaScript, TypeScript, HTML/CSS, SQL, Scala, VHDL
Libraries/Frameworks: React, Node.js, Express, NextJS, FastAPI, SpringBoot, NestJS, PyTorch, Pandas
Developer Technologies: Git, Supabase, Postman, Docker, Kubernetes, Azure, AWS, Swagger, Vercel, Railway